Differential Depth Response in Recursive Autoregressive Generation: Understanding Distributional Dynamics Beyond Binary Detection

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Abstract

Web-scraped training corpora increasingly contain LLM-generated text of unknown provenance depth—text that has been recursively generated through multiple autoregressive passes. Current detection methods provide only binary real-versus-synthetic classification, conflating qualitatively different material along the generation chain. We formulate *generation depth classification* (gen-0/gen-1/gen-2+) and characterize **differential depth response** (DDR): central-tendency features (entropy mean, TTR, hapax ratio) drift monotonically with depth ($|\rho| > 0.60$), while distributional-shape features (tail decay, entropy max) remain near-stationary ($|\rho| < 0.10$). Across 8 base-model generators from 5 families (0.6B–8B parameters), a lightweight 7-feature ensemble achieves leak-free binary F1 = 0.694 (95% CI [0.688, 0.700]) under leave-one-generator-out (LOGO) evaluation, outperforming existing detector baselines repurposed as ordinal predictors (best single: Avg-LogProb 0.765 3-class, $\Delta = -3.3$ pp). We further show that depth-aware data curation is statistically equivalent to random subsampling for downstream causal language model quality (TOST $p = 1.1 \times 10^{-4}$ at $\pm 2\%$ margin), establishing that distributional shift under recursive generation does not entail quality degradation—challenging implicit assumptions in the model collapse literature. The depth signal degrades substantially under instruction tuning and at high temperature, restricting the current scope to base models at moderate temperatures.

1 Introduction

Large language models increasingly generate text that re-enters training pipelines through web scraping, creating recursive feedback loops. Estimates suggest 7–17% of recent scientific publications contain LLM-generated content, and this proportion is growing. Understanding the provenance of such text—specifically, *how many autoregressive passes* separate it from human-authored source material—is important for characterizing the distributional dynamics of these feedback loops.

Binary detection methods (FastDetectGPT, Binoculars, LD-Score) classify text as human or machine-generated but cannot distinguish first-generation output (gen-1), which retains much of the source distribution's lexical properties, from deeper generations (gen-2+), which exhibit measurably reduced vocabulary diversity and compressed perplexity. This conflation obscures important distributional structure: gen-1 text may be nearly indistinguishable from human writing on surface metrics, while gen-4+ text shows substantial vocabulary impoverishment.

We formulate *generation depth classification* as a new task and introduce the **differential depth response** (DDR) framework for characterizing how distributional features evolve across recursive generation steps. Our central finding is that features split into two behavioral categories: *drifting* features (hapax ratio, TTR, entropy mean) that exhibit strong monotonic trends with depth, and *stationary* features (tail decay, entropy max) whose distributions remain approximately invariant. This split is consistent with recursive autoregressive generation inducing a location shift in the next-token log-likelihood (NLL) distribution while approximately preserving its shape.

We make three contributions:

1. **Task formulation & DDR characterization**: Generation depth classification beyond binary detection, with empirical validation of the drifting/stationary dichotomy across 8 generators from 5 model families (0.6B–8B).
2. **Lightweight depth classifier**: A 7-feature XGBoost ensemble achieving leak-free binary F1 = 0.694 under 8-generator LOGO, using a single 410M reference model.
3. **Negative finding**: Depth-aware data curation provides no benefit over random subsampling for downstream CLM quality, establishing depth fingerprinting as a diagnostic rather than curative tool.

2 Related Work

AI text detection. Perplexity-based detectors leverage statistical properties of model-generated text. DetectGPT [1] uses log-probability curvature; Fast-DetectGPT [2] replaces perturbation with conditional sampling; Binoculars [3] contrasts two models' cross-entropy; GLTR [4] visualizes token-level statistics. All operate in a binary detection paradigm and do not address generation depth.

Model collapse & iterative generation. Shumailov et al. (2024) show that training on model-generated data leads to distributional collapse. Dohmatob et al. (2024) provide theoretical analysis of iterative retraining dynamics. Briesch et al. (2025) study paraphrasing chains as "LLM as Broken Telephone," finding attractor convergence within 3–5 iterations. These works characterize population-level collapse but do not address sample-level depth estimation.

Data quality & provenance. Recent work on data curation (Penedo et al., 2024; Wettig et al., 2024) focuses on filtering low-quality text but does not distinguish generation depth. Our downstream utility experiment directly tests whether depth information improves curation, finding no benefit.

Vocabulary diversity & stylometry. Type-token ratio (TTR), hapax legomena ratio, and lexical diversity metrics have long been used in authorship attribution. We show these surface features capture substantial depth signal (F1 = 0.763 surface-only) but reference model features provide complementary information.

3 Methodology

3.1 Problem Formulation

Given a text passage x, we seek to classify its *generation depth* $d \in \{0, 1, 2+\}$, where $d = 0$ denotes human-authored seed text, $d = 1$ denotes first-pass LLM generation from the seed, and $d \geq 2$ denotes recursively generated text. The primary evaluation metric is *leak-free binary F1*: F1 for the gen-1 vs. gen-2+ distinction, excluding gen-0 to avoid seed-sharing leakage artifacts.

3.2 Generation Pipeline

We construct non-paraphrastic continuation chains: starting from human-authored seed texts (Wikipedia), each depth step feeds the previous output as a continuation prompt to the generator. We generate chains of depth 0–4 using 8 base-model generators from 5 families: Pythia (1B, 2.8B), OLMo-2 (1B), Qwen3 (0.6B, 1.7B, 8B), Llama-3.2 (3B), and Mistral-7B-v0.3.

3.3 Differential Depth Response (DDR) Features

We extract 7 features per text, computed relative to a reference model (Pythia-410M):

Table 1: Per-feature depth trend (linear regression, depths 0–4, Pythia-2.8B).

Feature	Category	Slope	R^2	Spearman ρ
hapax_ratio	Drifting	−0.063	0.88	−0.63
TTR	Drifting	−0.041	0.82	−0.61
entropy_mean	Drifting	−0.089	0.79	−0.60
entropy_std	Ambiguous	−0.031	0.41	−0.20
entropy_min	Stationary	−0.018	0.22	−0.15
tail_decay	Stationary	+0.003	0.04	+0.04
entropy_max	Stationary	+0.008	0.07	+0.05

Surface features (2). *Type-token ratio* (TTR): ratio of unique tokens to total tokens. *Hapax ratio*: fraction of tokens occurring exactly once.

Reference features (5). Given the reference model's next-token NLL distribution over the text: *entropy mean*, *entropy std*, *entropy min*, *entropy max*, and *tail decay* (rate parameter of exponential fit to the upper tail of the NLL distribution).

DDR dichotomy. We characterize each feature's depth trend via Spearman ρ against depth. Features with $|\rho| > 0.50$ are classified as *drifting*; those with $|\rho| < 0.10$ as *stationary*. This split reflects a location shift in the NLL distribution under recursive generation: central-tendency statistics (mean, diversity) shift monotonically, while shape statistics (tail behavior, extrema) are approximately preserved.

3.4 Classifier

We use XGBoost with 5-fold cross-validation for within-distribution evaluation and leave-one-generator-out (LOGO) for cross-generator generalization. LOGO trains on 7 generators and evaluates on the held-out 8th, averaging across all 8 folds.

4 Experimental Setup

Data. We generate chains from 1,000 Wikipedia seed passages per generator, with 5 depth levels (0–4) and default temperature $t = 0.7$ (nucleus sampling, $p = 0.95$). Total dataset: ∼37,500 records across 8 generators. Texts shorter than 50 tokens after generation are discarded.

Evaluation protocol. Primary metric: leak-free binary F1 (gen-1 vs. gen-2+) under LOGO. Secondary: 3-class macro F1 (gen-0/gen-1/gen-2+) and leave-one-family-out (LOFO) binary F1. All confidence intervals are bootstrap 95% CIs (10,000 resamples).

Baselines. We repurpose four existing detection methods as ordinal depth predictors: FastDetectGPT, Binoculars, LD-Score, and Avg-LogProb (average log-probability under the reference model). Each detector's continuous score is used as a single feature in the same XGBoost LOGO pipeline.

5 Results

5.1 DDR Characterization and Feature Ablation

Table 1 reports per-feature depth trends computed via linear regression across depths 0–4 on the Pythia-2.8B subset (∼4,600 records).

Table 2: Feature ablation: 5-fold CV macro F1 (Pythia-2.8B, ∼4,600 records).

Feature Set	Macro F1	95% CI	Δ vs Full
Full (7 features)	0.820	[0.807, 0.833]	—
Surface-only (TTR + hapax)	0.763	[0.747, 0.774]	−5.7 pp
Reference-only (5 entropy)	0.719	[0.706, 0.733]	−10.1 pp

Table 3: Leave-one-generator-out (LOGO) binary F1 (gen-1 vs. gen-2+).

Held-out Generator	Family	Size	Binary F1	95% CI
Pythia-1B	Pythia	1.0B	0.724	[0.707, 0.741]
Pythia-2.8B	Pythia	2.8B	0.711	[0.693, 0.729]
Qwen3-0.6B	Qwen	0.6B	0.702	[0.682, 0.722]
Qwen3-1.7B	Qwen	1.7B	0.698	[0.678, 0.718]
Llama-3.2-3B	Llama	3.0B	0.688	[0.668, 0.708]
OLMo-2-1B	OLMo	1.5B	0.679	[0.658, 0.700]
Qwen3-8B	Qwen	8.0B	0.647	[0.624, 0.670]
Mistral-7B-v0.3	Mistral	7.0B	0.633	[0.609, 0.657]
Average	—	—	**0.694**	[0.688, 0.700]

Three features exhibit strong monotonic drift with depth ($|\rho| \geq 0.60$, $R^2 \geq 0.79$): hapax ratio, TTR, and entropy mean. Two shape features show near-zero correlation ($|\rho| < 0.10$): tail decay and entropy max. Entropy std falls in an ambiguous zone ($\rho = -0.20$, $R^2 = 0.41$), suggesting a continuous rather than strictly bimodal response spectrum.

Table 2 reports feature ablation results.

The full 7-feature set outperforms both subsets (McNemar $p \approx 0$ for both comparisons), confirming complementarity. Surface features alone retain 93% of the full signal (0.763/0.820), indicating that vocabulary diversity metrics capture the majority of depth information. Reference features contribute an additional 5.7 pp, primarily improving gen-1/gen-2+ discrimination.

5.2 Cross-Generator Generalization

Table 3 reports LOGO binary F1 across all 8 generators (∼37,500 records total).

The average LOGO binary F1 is 0.694, with a per-generator range of 9.1 pp (Pythia-1B 0.724 to Mistral-7B 0.633). Performance degrades with model scale: the two largest generators (Mistral-7B at 0.633, Qwen3-8B at 0.647) yield the lowest F1, consistent with larger models compressing inter-depth feature margins. 86% of binary classification errors are false negatives (gen-2+ misclassified as gen-1), indicating systematic under-detection of deeper generations.

Table 4 shows leave-one-family-out (LOFO) results, confirming cross-family transfer with a modest 0.8 pp average degradation.

5.3 Comparison with Existing Detectors

Table 5 compares the 7-feature ensemble against existing detectors repurposed as ordinal depth predictors under the same LOGO protocol.

The 7-feature ensemble significantly outperforms all single detectors in 3-class F1 ($p < 0.001$). However, the margin over Avg-LogProb is modest (3.3 pp), and a multi-detector ensemble (4 detectors + surface features) achieves near-identical performance (3-class 0.795 vs. 0.798; binary 0.708 vs. 0.705, $p = 0.72$). The wide confidence intervals for single detectors (e.g., Avg-LogProb CI spanning 0.616–0.820) reflect high cross-generator variance.

Table 4: Leave-one-family-out (LOFO) binary F1.

Held-out Family	Binary F1	Δ vs LOGO
Pythia	0.668	−2.6 pp
Qwen	0.672	−2.2 pp
OLMo	0.714	+2.0 pp
Llama	0.701	+0.7 pp
Mistral	0.673	−2.1 pp
Average	**0.686**	**−0.8** pp

Table 5: Ordinal depth classification: 7-feature vs. existing detectors (LOGO). Bootstrap p tests 7-feature vs. each baseline.

Method	3-class F1	95% CI	Binary F1	p
7-feature	**0.798**	[0.786, 0.799]	**0.705**	—
Avg-LogProb	0.765	[0.616, 0.820]	0.695	<0.001
LD-Score	0.721	[0.588, 0.770]	0.672	<0.001
FastDetectGPT	0.627	[0.511, 0.762]	0.601	<0.001
Binoculars	0.579	[0.405, 0.664]	0.558	<0.001
4-det + surface	0.795	[0.781, 0.808]	0.708	0.72

5.4 Cross-Model Chains

Table 6 compares single-model chains (same generator at every depth step) with cross-model chains (different generator per step).

Cross-model chains yield substantially higher F1 (+7.0 pp binary, +13.6 pp LOGO), as generator switching across depth steps introduces additional distributional shifts beyond the depth signal alone. The single-model setting isolates the pure depth signal and is the more conservative evaluation.

5.5 Downstream Utility (Negative Finding)

Table 7 reports downstream causal language model perplexity when fine-tuning OLMo-1B on depth-stratified data subsets (3 random seeds, sample-count-matched).

Depth-aware filtering provides no advantage over random subsampling at equal sample count: depth 0+1 (ppl = 2.387) and binary-filtered (2.389) are statistically equivalent to size-matched random (2.367) under TOST at ±2% margin ($p = 1.1 \times 10^{-4}$). The ranking *all* < *size-matched* < *depth 0+1* ≈ *binary filtered* ≪ *depth 0 only* indicates that more data is always better, and depth-based selection slightly hurts by discarding useful training signal.

This negative finding challenges an implicit assumption in the model collapse literature: that distributional shift under recursive generation necessarily entails quality degradation. The DDR-characterized drift (vocabulary impoverishment, entropy compression) does not translate to worse downstream training outcomes at these depth levels.

6 Analysis

6.1 Depth Resolution Limits

Table 8 evaluates finer-grained 5-class ordinal classification (gen-0 through gen-4).

5-class F1 drops to 0.498, and ordinal methods (CORAL, Frank-Hall) provide no improvement, indicating that depth signal does not possess ordinal structure beyond coarse binning. Per-depth recall analysis reveals the structure: gen-0 recall is high (0.92) but gen-2/3/4 are

Table 6: Cross-model vs. single-model chain classification.

Setting	Binary F1	3-class F1	LOGO Binary F1
Single-model	0.694	0.762	0.694
Cross-model	0.764	0.789	0.830
Δ	+7.0 pp	+2.7 pp	+13.6 pp

Table 7: Downstream CLM perplexity by training data depth composition (3 seeds, OLMo-1B).

Variant	Train Size	PPL (mean ± std)	Δ vs Size-matched
All depths	12,299	2.335 ± 0.0003	—
Size-matched random	3,951	2.367 ± 0.0003	ref
Depth 0+1 only	3,951	2.387 ± 0.0004	+0.020
Binary filtered	3,951	2.389 ± 0.0006	+0.022
Depth 0 only	797	2.681 ± 0.0006	+0.314

poorly separable (recall 0.29–0.41, inter-class confusion 48–51%). The depth signal is effectively binary-plus-one: source-adjacent (gen-0/1) versus deep (gen-2+), with limited resolution within either group.

6.2 Scale Effects

Table 9 examines how model scale affects depth signal strength.

Qwen3-8B shows a 10.6 pp degradation in LOGO binary F1 relative to the 6-generator average, with feature drift magnitudes compressed to ∼72% of smaller models. The DDR pattern (drifting > stationary separation) is preserved, but weakened. This suggests that higher-capacity models produce more uniform output across depth levels, compressing the inter-depth distributional margins that DDR exploits.

6.3 Reference Model Independence

Table 10 tests whether DDR patterns persist across different reference models.

Drift direction is consistent across all reference models (all entropy_mean $\rho < 0$, all shape features $|\rho| < 0.10$), confirming that DDR is not an artifact of the specific reference model. However, drift *magnitude* varies substantially: OLMo reference produces only ∼55% of Pythia's drift magnitude. LOGO F1 remains comparable (0.803–0.816), indicating that classification performance is robust to reference model choice even as the underlying DDR characterization changes in scale.

7 Conclusion

We introduced generation depth classification and characterized differential depth response (DDR) across 8 base-model generators from 5 families. Central-tendency features drift monotonically with depth while shape features remain near-stationary, consistent with recursive generation inducing a location shift in the NLL distribution. A lightweight 7-feature ensemble achieves leak-free binary F1 = 0.694 under LOGO (per-generator range: 0.633–0.724), modestly outperforming single detector baselines but matching multi-detector ensembles. The depth signal is effectively limited to 3-class resolution and degrades with model scale.

Our negative downstream finding—depth-aware curation is equivalent to random subsampling for CLM quality—suggests that the distributional shifts characterized by DDR do not

Table 8: Ordinal depth classification resolution.

Granularity	Method	Macro F1
3-class	XGBoost	0.762
5-class	XGBoost	0.498
5-class	CORAL	0.502
5-class	Frank-Hall	0.491

Table 9: Scale effect: depth signal degradation with generator size.

Generator	Size	LOGO Binary F1	hapax ρ	entropy_mean ρ
Pythia-1B	1.0B	0.724	-0.65	-0.60
Qwen3-1.7B	1.7B	0.698	-0.63	-0.60
Pythia-2.8B	2.8B	0.711	-0.63	-0.60
Qwen3-8B	8.0B	0.647	-0.56	-0.48
Trend	—	**Declining**	**Compressing**	

entail quality degradation at the depth levels studied. This challenges the assumption that recursive generation necessarily produces inferior training data and positions DDR as a diagnostic framework for understanding distributional dynamics rather than a tool for data curation.

Limitations. Three key limitations bound the scope of this work. (1) *Base models only*: instruction-tuned models show substantially degraded depth signal (LOGO F1 drops to 0.338 for OLMo-Instruct vs. 0.763 for OLMo-Base), restricting applicability to base model outputs—a shrinking fraction of deployed LLM-generated text. (2) *Temperature sensitivity*: reliable performance is restricted to $t \in [0.5, 0.9]$; at $t = 1.0$, entropy mean ρ drops to near-zero and cross-temperature transfer degrades severely. (3) *Scale degradation*: larger generators (7B+) compress depth signal, and the observed trend suggests further degradation at frontier model scales.

A Prompt Type Sensitivity

Table 11 compares three prompt types: continuation, elaboration, and inspired rewriting (Pythia-2.8B, 1000 seeds).

Continuation chains yield the strongest within-type signal (CV F1 = 0.781), while elaboration and rewriting show higher depth retention but weaker depth discrimination ($\Delta = 5.1$ pp). Cross-prompt-type transfer (LOPTO) averages 0.753, indicating partial but imperfect generalization.

B Paraphrastic vs. Non-Paraphrastic Chains

Table 12 compares paraphrastic and non-paraphrastic (continuation) chains.

Paraphrastic chains exhibit attractor convergence (depth retention 0.992), reducing effective depth classes to $\sim$2. The binary F1 gap ($\Delta = 3.3$ pp) is moderate, suggesting the depth signal is a generic property of recursive generation rather than unique to non-paraphrastic chains, though non-paraphrastic chains sustain an additional class of depth resolution.

C Multi-Domain Evaluation

Table 13 reports per-domain binary F1 under LOGO evaluation.

Table 10: Cross-reference-model DDR comparison.

Reference Model	Size	LOGO F1	entropy_mean ρ	Drift range
Pythia-410M	410M	0.803	-0.60	$[-1.19, -1.40]$
OLMo-2-1B	1.5B	0.807	-0.50	$[-0.61, -0.91]$
Qwen3-0.6B	0.6B	0.816	-0.56	$[-0.96, -1.35]$

Table 11: Per-prompt-type classification and drift statistics (Pythia-2.8B).

Prompt Type	CV F1	LOPTO F1	Depth Ret.	ent_mean ρ	hapax ρ	TTR ρ
Continuation	0.781	0.762	0.831	-0.557	-0.740	-0.629
Elaboration	0.730	0.762	0.986	-0.471	-0.828	-0.666
Rewriting	0.737	0.734	0.939	-0.477	-0.800	-0.659

Performance is strongly domain-dependent: Wikipedia (primary training domain) achieves 0.815, while academic text—the domain where LLM contamination is most frequently cited—yields only 0.628 ($\Delta = -18.7$ pp). Cross-domain LOGO drops to 0.668.

D Robustness to Human Edits

Table 14 evaluates robustness under simulated human editing.

Light editing (5% random replacement or LLM rephrasing) reduces binary F1 by 4.5 pp. Degradation is approximately linear with perturbation intensity, suggesting the depth signal resides in surface-level distributional properties susceptible to text modification.

E LD-Score Feature Extension

Table 15 evaluates the addition of LD-Score (character-distribution signature) as an 8th feature.

LD-Score alone achieves 3-class F1 = 0.721, but its marginal contribution when added to the 7-feature set is negligible (0.4 pp), indicating substantial information overlap with existing features.

F End-to-End QLoRA Classifier

Table 16 compares the 7-feature ensemble with end-to-end QLoRA fine-tuned classifiers.

QLoRA classifiers substantially outperform the 7-feature ensemble (+12.6 to +17.3 pp binary F1), indicating the text contains substantially more depth signal than the 7-feature set captures. The 7-feature ensemble trades absolute performance for 47× lower inference cost. Notably, instruct-model QLoRA classifiers achieve the highest F1, suggesting that instruction tuning degrades the specific features selected rather than the depth signal itself.

G Text Length Control

Table 17 controls for text length as a potential confound.

Length-depth correlation is weak ($\rho = 0.12$, $p = 0.001$). Length-matched subsampling reduces binary F1 by 4.6 pp (0.770 → 0.724), indicating that approximately 6% of the signal is attributable to text length. Feature depth trends (TTR, hapax ratio) are preserved after length normalization, confirming that the core DDR pattern is not a length artifact.

Table 12: Paraphrastic vs. non-paraphrastic chain comparison (Pythia-2.8B).

Chain Type	CV F1	Binary F1	Depth Ret.	ent_mean ρ	Eff. Classes
Continuation	0.781	0.694	0.831	-0.557	3
Paraphrastic	0.718	0.661	0.992	-0.390	$\sim$2
Δ	+6.3 pp	+3.3 pp	—	—	+1

Table 13: Per-domain binary F1 (LOGO across generators).

Domain	Seeds	Binary F1	95% CI	Δ vs Wiki
Wikipedia	500	0.815	[0.796, 0.834]	—
News	500	0.712	[0.690, 0.733]	-10.3 pp
Code	500	0.687	[0.664, 0.710]	-12.8 pp
Academic	500	0.628	[0.603, 0.652]	-18.7 pp
Cross-domain LOGO	2000	0.668	[0.654, 0.682]	—

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Table 14: Robustness to simulated human edits.

Perturbation	Binary F1	Δ vs Baseline	Edit Distance
None (baseline)	0.695	—	0.0
Random 5%	0.650	−4.5 pp	∼0.05
Random 10%	0.614	−8.1 pp	∼0.10
Random 20%	0.591	−10.4 pp	∼0.20
LLM rephrase	0.647	−4.5 pp	0.507

Table 15: LD-Score feature addition (3-class LOGO).

Feature Set	LOGO F1	Δ
7 features (baseline)	0.798	—
7 + LD-Score	0.802	+0.4 pp
LD-Score only	0.721	—

Table 16: End-to-end (QLoRA) vs. 7-feature ensemble.

Method	Model	3-class F1	Binary F1	Cost
7-feature XGBoost	—	0.762	0.694	1×
QLoRA	OLMo-Instruct	0.888 ± 0.007	0.836 ± 0.010	47×
QLoRA	Qwen-Instruct	0.907 ± 0.008	0.867 ± 0.014	47×
QLoRA	OLMo-Base	0.859 ± 0.004	0.793 ± 0.010	47×
QLoRA	Qwen-Base	0.904 ± 0.009	0.860 ± 0.012	47×

Table 17: Length control analysis.

Condition	Binary F1	Δ vs Unmatched
Unmatched (original)	0.770	—
Length-normalized features	0.748	−2.2 pp
Length-matched subsample	0.724	−4.6 pp